

Project Charter



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PROJECT PROPOSAL

Nexscholar System

Faculty of Computing UTM

Team Name: **Team 6**

Team Members:

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1. Introduction

NextScholar is a research discovery platform that aims to connect various people from different academic levels, including students, lecturers, supervisors, grantees, and collaborators, through AI-inferred preferences [1]. This platform uses AI to recommend suitable research opportunities and academic connections according to the users' research interests, experiences, and preferences. These services are provided by a centralized platform for research collaboration and communication. NextScholar tries to simplify the process of finding suitable partners by focusing on the academic interaction between its users.

Despite providing AI-supported research discovery and matching capabilities, several usability and functionality problems were detected during meetings and workshops that were conducted with the stakeholders. Currently, lecturers face challenges while searching for and evaluating potential research students because their available profile information may not be enough for quick comparison and decision-making. Lecturers mostly need to reach out to access multiple profile pages to gather the information required to determine whether a student is suitable for a research project. Moreover, students are struggling with selecting supervisors whose interests and experiences satisfy students' academic goals and preferences.

Besides that, one of the important issues is limited personalization of the current matching process, which leads to minimizing the accuracy and suitability of the recommendations that are created by AI. Stakeholders suggested collecting extra information, such as research preferences, experience, and interests, during the sign-up process. Additionally, users would benefit from the quick research suggestions that are based on the profile, hybrid AI and keyword-based search functionality, enhanced profile previews, and direct links to communication platforms such as WhatsApp, Telegram, and LinkedIn to facilitate easier collaboration and communication.

The workshop also highlighted the useability challenges related to the current tutorial and onboarding experience. Existing guidance is mainly text-based. According to that, users may not fully understand the features of the platform during their initial interaction with the system. Stakeholders also suggested adding interactive tutorial pop-ups, guided onboarding sessions, video demonstrations, visual walkthroughs, and personalized tutorials based on users' interests and objectives. These enhancements help in reducing confusion, improving feature use, and providing an easier-to-use experiment for both new and current users.

The primary stakeholders of the NexScholar platform are students, lecturers, and supervisors. Students seek suitable supervisors and research opportunities that align with their interests and academic goals. Lecturers and supervisors require efficient tools to identify, evaluate, and communicate with potential research students. Therefore, this project proposes several enhancements focusing on AI matching, interactive tutorials, and user engagement features. These improvements aim to provide more accurate recommendations, simplify research partner discovery, improve onboarding experiences, increase user engagement, and ultimately make NexScholar a more effective platform for academic collaboration and research discovery.

Figure 1.1 shows the NexScholar homepage, which emphasizes research collaboration and AI-driven networking features.

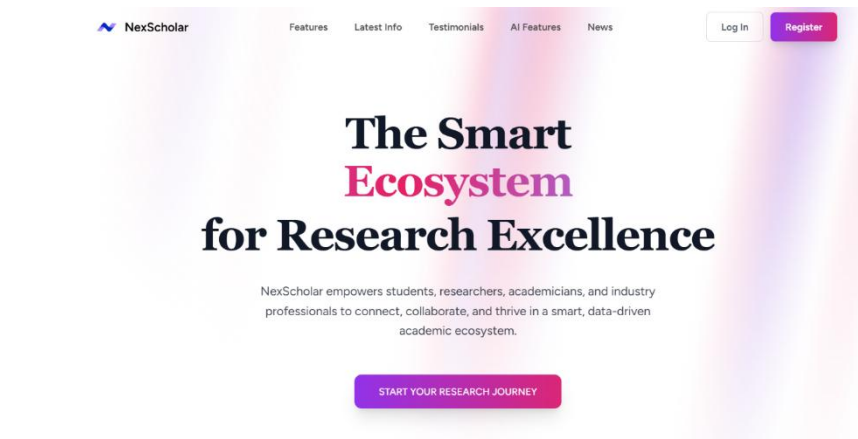


Figure 1.1: NexScholar Homepage [2]

Figure 1.2 shows the registration page where new users can create an account by providing their username, email address, and password before accessing the platform features.

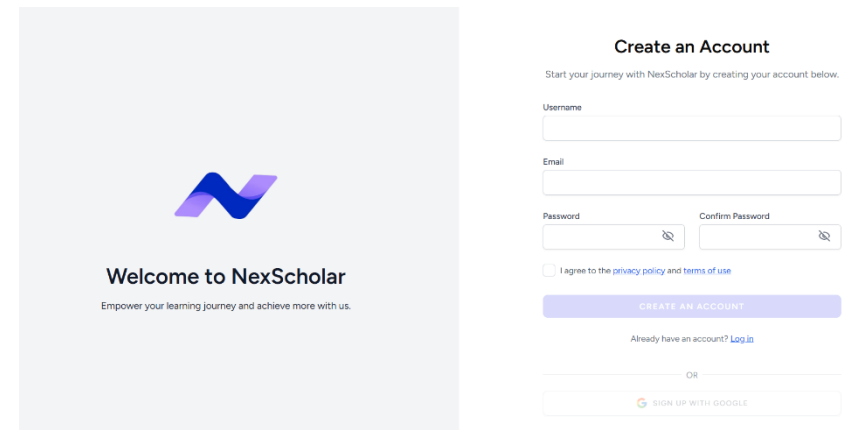


Figure 1.2: NexScholar Registration Page [2]

Figure 1.3 illustrates the onboarding questionnaire used to collect users' academic interests and preferences, enabling the platform to provide more personalized recommendations

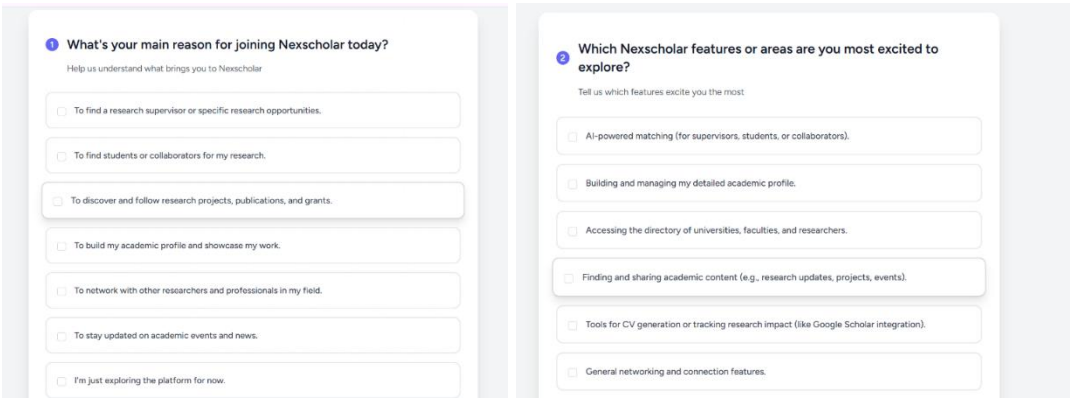


Figure 1.3: NexScholar Onboarding Questionnaire [2]

2. Existing Systems

There are many platforms that offer features for academic networking, profile management, and research discovery. In this project, three systems were chosen to compare them, including LinkedIn, ResearchGate and Google Scholar. These systems were chosen because they offer professional networking, researcher discovery, and academic collaboration, like some of the features in NexScholar. However, they do not fully offer the AI-based student-supervisor matching and interactive onboarding features suggested in this project.

Table 2.1: Comparison of existing systems

Features	LinkedIn [3]	ResearchGate [4]	Google Scholar [5]	NexScholar
User Profile Management	Yes	Yes	Yes	Yes
Academic Research Profile	Limited	Yes	Yes	Yes
Research Interest Display	Yes	Yes	Limited	Yes
Research Collaboration Discovery	Limited	Yes	Limited	Yes
AI-Based Recommendation	Yes	Limited	Yes	Yes
Advanced Search Function	Yes	Yes	Yes	Yes
Keyword-Based Search	Yes	Yes	Yes	Yes
Research Opportunity Discovery	Limited	Limited	No	Yes
Supervisor/Student Matching	No	No	No	Yes
External Communication Links	Yes	Limited	No	Yes
Personalized User Recommendations	Yes	Limited	Limited	Yes
Interactive Tutorial Support	Limited	No	No	Yes
User Engagement Features	Yes	Limited	No	Yes
Academic Networking	Yes	Yes	Limited	Yes
Mobile Accessibility	Yes	Yes	Limited	Yes
Research Publication Management	No	Yes	Yes	Yes

According to table 2.1, Although platforms such as LinkedIn, ReseachGate and Google Scholar provide a strong support for academical communication and research discovery, none of them fully support AI-based student-supervisor matching and personalized onboarding features. This shows the need for an integrated platform such as NexScholar that combines intelligent recommendation systems with enhanced collaboration and onboarding functionalities.

3. Epic, User Story, Product Backlog and Sprint Backlog

According to the discussions with stakeholders, four major epics were identified: AI Matching & Search, Research Discovery & Collaboration, Interactive Tutorial System, and Personalized Onboarding. These epics represent the primary improvements to enhance the NexScholar user experience and improve research matching effectiveness. Table 3.1 presents the product backlog, including the identified epics, user stories, priorities, and story point estimations for each backlog item.

Table 3.1. Product Backlog Item

EPIC	ID	User Story	Priority	Estimation
E1: AI Matching & Search	US101	As a student, I want to provide my research interests, preferences, and experiences during registration so that I can receive more accurate research recommendations.	High	5
	US102	As a user, I want quick search suggestions based on my profile so that I can find relevant research opportunities faster.	High	8
	US103	As a user, I want to switch between AI-based search and keyword-based search so that I can use my preferred search method.	High	5
E2: Research Discovery & Collaboration	US201	As a lecturer, I want to view and evaluate potential students efficiently so that I can identify suitable research candidates.	High	5
	US202	As a lecturer, I want to view a profile preview popup containing key information so that I can compare candidates quickly.	Medium	3
	US203	As a user, I want direct links to WhatsApp, Telegram, and LinkedIn profiles so that I can communicate with collaborators easily.	Low	2
E3: Interactive Tutorial System	US301	As a new user, I want tutorial pop-ups when using the system for the first time so that I can understand platform features quickly.	High	8
	US302	As an existing user, I want tutorials for newly introduced features so that I can learn system updates easily.	Medium	3
	US303	As a user, I want video and visual tutorials so that I can learn more effectively.	High	8
E4: Personalized Onboarding	US401	As a new user, I want personalized onboarding questions based on my interests so that the platform can provide relevant recommendations.	High	8
	US402	As a user, I want tutorial recommendations based on my selected interests so that I can learn relevant features only.	Medium	5

Table 3.2 presents the sprint backlog, including the user stories and their corresponding technical task breakdowns required to implement the proposed system enhancements.

Table 3.2 Sprint Backlog

ID	User Story	Technical Task Breakdown
101	Collect research interests during registration	Design onboarding form, update database schema, implement validation, testing
102	Profile-based quick search suggestions	Develop recommendation logic, connect profile data to search module, testing
103	AI Search / Keyword Search Toggle	Design search toggle interface, implement AI search mode, implement keyword search mode, testing
201	Student evaluation and discovery	Design popup UI, retrieve profile data, testing
202	Profile preview popup	Design profile preview popup interface, display key profile information, implement hover/click functionality, testing
203	External communication links	Add external communication fields, validate links, testing
301	New user tutorial popup	Design onboarding workflow, create tutorial popups, testing
302	New feature tutorial popup	Implement update notification mechanism, testing
303	Video and visual tutorials	Develop multimedia tutorial pages, integrate video support, testing
401	Personalized onboarding questions	Create preference questionnaire, store user preferences, testing
402	Tutorial recommendation feature	Develop recommendation rules, connect tutorial database, testing

4. Team roles values, working arrangements, rules, and norms

Table 4.1: Team Role

Role	Person in charge	Responsibilities
Product Owner	Leen Milhem	Gather stakeholder requirements, manage product backlog, prioritize user stories, ensure project objectives meet stakeholder needs.
Scrum Master	Mohammad Munther	Facilitate Scrum meetings, monitor project progress, remove obstacles, ensure Agile practices are followed by the team.
Development Team	Shadin Hanafi, Nour Gadallah, Leen Milhem, Mohammad Munther	Design solutions, create prototypes, implement project features, prepare documentation, perform testing and validation.

Table 4.2: Team Rule

Rule for as a Team	Description
Preparedness for Meetings	Team members must attend meetings prepared with completed tasks, updates, and questions for discussion.
Effective Communication	Team members should respond to messages within 24 hours and inform the team early if they encounter problems or delays.
Respect Deadlines	All assigned tasks must be completed according to the agreed schedule to avoid affecting sprint progress.
Active Participation	Every member must contribute ideas, feedback, and support during discussions and sprint activities.
Maintain Work Quality	Team members should review their work before submission to ensure accuracy and consistency.

5. Project Cost Estimation

The NexScholar project uses Agile Story Point estimation based on the Fibonacci sequence (2, 3, 5, 8) to reflect the relative complexity and effort required for each user story [6]. Story Points are assigned by the development team through Planning Poker [7], considering effort, complexity, and uncertainty. The total estimated effort for this project is 60 Story Points across 11 user stories, as detailed in Table 5.1 below.

Table 5.1: Project Cost Estimation using Story Points

EPIC	ID	User Story	Story Points	Priority
E1	US101	Research preferences during registration	5	High
	US102	Profile-based quick search suggestions	8	High
	US103	AI / Keyword search toggle	5	High
E2	US201	Student evaluation and discovery	5	High
	US202	Profile preview popup	3	Medium
	US203	External communication links	2	low
E3	US301	First-time user tutorial popup	8	High
	US302	New feature tutorial popup	3	Medium
	US303	Video and visual tutorials	8	High
E4	US401	Personalized onboarding questions	8	High
	US402	Tutorial recommendations	5	Medium
Total Story Points		11 User Stories across 4 Epics	60	-

The NexScholar Enhancement Project consists of 11 user stories grouped into four epics with a total estimated effort of 60 story points. The highest effort is allocated to AI Matching & Search and Personalized Onboarding features because they involve recommendation logic, user preference analysis, and integration of multiple system components. These estimations will assist the team in sprint planning, task prioritization, and workload distribution throughout the project lifecycle.

6. Project Planning and Schedule

The project planning and schedule for the NexScholar Enhancement Project were developed based on the Hybrid Agile Model. The project is divided into five weeks, including project charter preparation, three Agile Scrum implementation sprints, and final submission activities. Tasks are assigned according to the responsibilities of each team member to ensure effective collaboration, progress monitoring, and timely completion of project deliverables. The planned activities, duration, and person in charge are presented in Tables 6.1 and 6.2 .

Table 6.1 Project Activities and Responsibilities

Activity	Tasks	Person In Charge	Week 11 30-May	Week 12 02–08 Jun	Week 13 08–14 Jun	Week 14 15–21 Jun	Week 15 22–28 Jun
Project Charter	Write & finalize all charter sections (1–7)	All Members	All Members				
Preparation & Submission	Review & submit charter document	Mohammed (Scrum Master)	Mohammed				
Sprint #1	Daily Standup Meeting & SCRUM deliverables	Mohammed (Scrum Master)		Mohammed			
Agile Scrum Implementation	Product Backlog, User Story & Functional Decomposition	Leen (Product Owner)		Leen			
	Sprint Deliverable: Video recording demo	Nour & Shadin (Dev Team)		Nour & Shadin			
	SPRINT DELIVERABLE SUBMISSION	All Members		All Members			
Sprint #2	Daily Standup Meeting & SCRUM deliverables	Mohammed (Scrum Master)			Mohammed		
Agile Scrum Implementation	Product Backlog, User Story & Functional Decomposition	Leen (Product Owner)			Leen		
	Initial low-fidelity prototype wireframe demo – Release 1	Nour & Shadin (Dev Team)			Nour & Shadin		
Sprint #3	Daily Standup Meeting & SCRUM deliverables	Mohammed (Scrum Master)				Mohammed	
Agile Scrum Implementation	SCRUM deliverables review	Leen (Product Owner)				Leen	
	Initial low-fidelity prototype wireframe demo – Release 2	Nour & Shadin (Dev Team)				Nour & Shadin	
Final Submission	Final wireframe submission & documentation	All Members					All Members
Low-fidelity Prototype							

Table 6.2: Gantt Chart for NexScholar Enhancement Project

Activity	Tasks	Person In Charge	Week 11 May 2026 30	Week 12 02- 08 May 2026	Week 13 08 Jun 2026 09-	Week 14 15-21 Jun 2026	Week 15 22-28 Jun 2026
Project Charter	Write & finalize all charter sections (1-7)	All Members					
Preparation & Submission	Review & submit charter document	Mohammed (Scrum Master)					
Sprint #1 Agile Scrum Implementation	Daily Standup Meeting & SCRUM deliverables	Mohammed (Scrum Master)					
	Product Backlog, User Story & Functional Decomposition	Leen (Product Owner)					
	Sprint Deliverable: Video recording demo	Nour & Shadin (Dev Team)					
	SPRINT DELIVERABLE SUBMISSION	All Members					
Sprint #2 Agile Scrum Implementation	Daily Standup Meeting & SCRUM deliverables	Mohammed (Scrum Master)					
	Product Backlog, User Story & Functional Decomposition	Leen (Product Owner)					
	Initial low-fidelity prototype wireframe demo — Release 1	Nour & Shadin (Dev Team)					
Sprint #3 Agile Scrum Implementation	Daily Standup Meeting & SCRUM deliverables	Mohammed (Scrum Master)					
	SCRUM deliverables review	Leen (Product Owner)					
	Initial low-fidelity prototype wireframe demo — Release 2	Nour & Shadin (Dev Team)					
Final Submission Low-fidelity Prototype	Final wireframe submission & documentation	All Members					

Based on the project schedule, the NexScholar Enhancement Project begins with project charter preparation and submission, followed by three Agile Scrum implementation sprints focusing on backlog development, user story refinement, and low-fidelity prototype creation. The assigned responsibilities ensure that each team member contributes according to their role, while daily stand-up meetings and Scrum deliverables support continuous monitoring and collaboration. The project concludes with the submission of the final prototype and supporting documentation, ensuring that all planned objectives are completed within the allocated timeline

7. References

- [1] *Project Write-up NexScholar Enhancement Project | Prepared for SCSE1203 Software Engineering Principles NexScholar Enhancement Project AI Matching, Interactive Tutorial, and User Engagement Features.* 2026.
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